

Aaryan Sharma

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Data Scientist with hands-on experience in machine learning, statistical inference, NLP, A/B testing, and time series forecasting across finance and marketing. Skilled in Python, SQL, and R -- building ETL pipelines and translating ambiguous business problems into statistically rigorous, production-ready insights using cloud and data engineering tools (AWS, Databricks, Snowflake).

TECHNICAL SKILLS

Languages & Tools: Python, SQL, R, Excel, Snowflake, Tableau, Docker, Git/GitHub, SPSS, QGIS, AWS EC2, Databricks

Methods: A/B Testing, Bayesian Methods, Data Engineering, Data Mining, Deep Learning, Explainable Boosting Machines (EBM), ETL & Data Pipelines, Financial Analytics, Fraud Analytics, Marketing Analytics, Machine Learning, NLP / Text Analytics, Optimization, Regression Methods, Risk & Simulation Analytics, Statistical Inference & Hypothesis Testing, Time Series Forecasting, Web Scraping

Certifications: AWS Certified, Google Analytics 4, Google Ads, Google Display Ads

EDUCATION

Master of Science in Analytics | NC State University, Institute for Advanced Analytics, Raleigh, NC

May 2025

Bachelor of Science in Economics, Business Administration | NC State University, Raleigh, NC

May 2024

Concentration in Marketing, Business Analytics

PROFESSIONAL EXPERIENCE

Capital One

Chicago, IL

Senior Data Science Analyst – Discover IT 5% Rewards Team

Aug. 2025 – Present

- Conducted NLP-based sentiment and topic analysis (BERTopic, NLTK, VADER) on 7,000 open-ended survey responses, engineering a new variable for the category optimization tool and presenting customer-preferred category insights to leadership
- Designed A/B testing experiments for email and site marketing campaigns across customer segments, driving an average 3.32% lift in Discover Cashback activation rates through winning variant selection
- Built a SARIMAX + Monte Carlo forecasting methodology and cut turnaround time by 71% for rewards sales and cost forecasting
- Led technical analysis for a new anchor spending-category launch, partnering with fraud and product teams to define scope and sizing; projected 2.4 MM increase in user growth and 4.3 B in incremental sales for quarterly promotion
- Built a unified KPI monitoring framework across two merged data systems (Capital One/Discover), engineering cross-stack SQL/Snowflake ETL pipelines that cut reporting time by 50% and eliminated manual error
- Analyzed transaction and enrollment data to evaluate consumer behavior around 5% promoted category sales, identifying category selection as a key driver of customer engagement, card retention, and spend.
- Investigated activation rate and rewards cost disparities between legacy and migrated Discover accounts, identifying the root cause to a small-sample statistical artifact rather than a true trend shifts

NC State University

Raleigh, NC

Teaching Assistant – Marketing Research, Predictive Analytics

Jan. 2024 – Present

- Provide academic support and mentorship to 300+ students in predictive analytics, statistical modeling, machine learning, and marketing research methodologies
- Guide students in creating business-related recommendations from in-depth data analysis from Qualtrics data
- Effectively translate supervised/unsupervised learning concepts to a non-technical audience

Bank of America

Charlotte, NC

Global Operations Development Analyst Intern – Global Payments

June 2023 – Aug. 2023

- Automated a GPO IT complaint tracker with Excel PowerPivot, reaching 92% audit compliance and cutting risk exposure 33%
- Built an ARIMAX to forecast invoice costs at 82% accuracy, supporting financial planning for ACH/WTX payment processing

VisionPoint Marketing

Raleigh, NC

Business Development Analyst

Jan. 2023 – Apr. 2023

- Leveraged descriptive (Summary Statistics) and predictive analytics (Time Series) in Excel to optimize inquiry processing, reducing response time by 25%
- Conducted data-driven research that secured three marketing-hunting licenses, improving lead generation for future business opportunities

PRACTICUMS

Wells Fargo

Sept. 2024 – Apr. 2025

- Identified KPI drivers for the complaints team to enhance customer service outcomes and improve KPI performance; evaluated driver impacts utilizing Python, statistical analysis, and machine learning
- Employed time series forecasting in R to analyze and predict demand fluctuations for the escalated complaints team, enabling better optimization of employee allocation
- Built predictive models using logistic regression to provide interpretable insights for stakeholders and leveraged Explainable Boosting Machine (EBM) to optimize predictions, gaining both global and local understanding of variable impacts on cases

- Engineered an outlier detection model in Python to identify customer service representatives exhibiting performance anomalies, generating alert reports in Tableau to notify stakeholders

The Agriculture Institute @ NC State

Jan. 2024 – Apr. 2024

- Deployed a python-based Streamlit K-Means/DBSCAN clustering dashboard flagging at-risk students for faculty outreach, lifting in-class engagement and student retention by 33%
- Built an interactive Tableau dashboard to provide real-time post-graduate salary insights, enhancing recruitment efforts for prospective students

East Bower Cider Company

Feb. 2024 – Apr. 2024

- Conducted consumer segmentation and text analysis in R, identifying customer groups and key sentiment drivers to optimize ad messaging and social media strategy, leading to a 38% increase in engagement
- Collaborated with a marketing team to design mock social media creatives and assess expected engagement impact using historical consumer interaction data

TSV Analytics

Aug. 2023 – Dec. 2023

- Optimized paid search campaigns using Google Ads, GA4, and SEO tools, leading to a 2700% increase in conversions by refining keyword strategy, bid adjustments, and audience segmentation
- Conducted A/B testing to enhance ad placements and keywords, improving user engagement and conversion rates

Safewell Essentials

Jan. 2022 – Apr. 2022

- Conducted segmentation analysis, optimizing marketing investment and conversion rates by 20%
- Designed consumer surveys and focus groups, applying statistical analyses (Paired t-tests, ANOVA) to translate raw feedback into actionable consumer insights.

EXTERNAL PROJECTS

- **Google Review Dashboard:** Engineered and deployed a production NLP pipeline (BS4, Selenium web scraping; VADER, BERTopic sentiment/topic modeling) containerized with Docker and CI/CD, analyzing customer sentiment and pricing trends from Google Reviews
- **McDonalds Sentiment Reviews:** Conducted large-scale text analysis using NLP techniques (NLTK, VADER, BERTopic), categorizing reviews across 20+ locations to assess brand sentiment and key topics
- **NC State Women's Tennis Dashboard:** Designed a Tableau and Python-based dashboard analyzing match performance data to help coaches tailor training plans for players
- **2024 NC State Plant Science Initiative Hackathon:** Leveraged feature selection techniques (PCA, Lasso Regression) and ML models to analyze 2,800+ agricultural variables, identifying key factors affecting corn yield.
- **2024 NC Education Datathon:** Performed geospatial analysis using QGIS and statistical tools to evaluate correlations between Career & Technical Education (CTE) and student achievement, providing data-driven policy recommendations to stakeholders.
- **2023 NC Education Datathon:** Applied linear and logistic regression models in R to analyze school performance trends and key drivers in standardized testing performance, transforming public education datasets into actionable insights